The universal property of the pushout: maps out of A sqcup^C B are the cocones under the span

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:pushout-ump' (source lines 953-957); the proof environment begins on the next line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own definition of a cocone under a span together with the canonical cocone of a pushout, because the statement is phrased with that notion and the proof's first round trip is an equality inside that type.

Editorial note. The retained counter values (chapter 6, section 8, theorem offset 0) make the retained cocone definition print as Definition 6.8.1 and the counted lemma as Lemma 6.8.2, matching the original numbering.

Editorial note. No retained line refers to any label, citation, figure, included file or non-ASCII character, so the package declares no external reference, no bibliography and no asset; this item compiles from the retained text alone.

Editorial scope. One original result of the HoTT Book, counted once: The universal property of the pushout: maps out of A sqcup^C B are the cocones under the span.

Allowed standard prerequisites (editorial).

- Higher inductive pushouts and their recursion principle. For a span C -f-> A, C -g-> B the pushout A sqcup^C B is generated by inl : A -> A sqcup^C B, inr : B -> A sqcup^C B and glue : Prd{c:C} inl(f(c)) = inr(g(c)); its recursion principle defines a map s : A sqcup^C B -> D from values on the three generators, with the computation rule map-func{s}(glue(c)) = the given path.. The inverse direction of the equivalence is defined exactly by this recursion principle at source lines 971-976, and the second round trip uses the computation rule for glue at source lines 1003-1005.
- Cocones under a span. A cocone under D = (A <-f- C -g-> B) with vertex D is a triple (i,j,h) with i : A -> D, j : B -> D and h : Prd{c:C} i(f(c)) = j(g(c)); the type of such cocones is written cocone(D,D).. The statement is an equivalence between A sqcup^C B -> E and cocone(D,E), and the retained context supplies the author's own definition of the latter.
- Function extensionality and paths between sigma-type elements. Two functions are equal as soon as they are homotopic; equality of triples

in a sigma type is componentwise.. The second round trip at source lines 996-1006 concludes equality of maps A sqcup^C B -> E after comparing values on inl, inr and glue.

Definition 6.8.1. Given a span $\mathscr{D} = (A \xleftarrow{f} C \xrightarrow{g} B)$ and a type D, a **cocone under $\mathscr{D}$ with vertex D** consists of functions $i : A \to D$ and $j : B \to D$ and a homotopy $h : \prod_{(c:C)}(i(f(c)) = j(g(c)))$:

$$\begin{array}{ccc} C & \xrightarrow{g} & B \\ {\scriptstyle f}\downarrow & \swarrow^{h} & \downarrow{\scriptstyle j} \\ A & \xrightarrow[i]{} & D \end{array}$$

We denote by $\mathsf{cocone}_{\mathscr{D}}(D)$ the type of all such cocones, i.e.

$$\mathsf{cocone}_{\mathscr{D}}(D) :\equiv \sum_{(i:A\to D)} \sum_{(j:B\to D)} \prod_{(c:C)} (i(f(c)) = j(g(c))).$$

Of course, there is a canonical cocone under $\mathscr{D}$ with vertex $A \sqcup^C B$ consisting of inl, inr, and glue.

$$\begin{array}{ccc} C & \xrightarrow{g} & B \\ {\scriptstyle f}\downarrow & \swarrow^{\mathsf{glue}} & \downarrow{\scriptstyle \mathsf{inr}} \\ A & \xrightarrow[\mathsf{inl}]{} & A \sqcup^C B \end{array}$$

The following lemma says that this is the universal such cocone.

Lemma 6.8.2. *For any type E, there is an equivalence*

$$(A \sqcup^C B \to E) \;\simeq\; \mathsf{cocone}_{\mathscr{D}}(E).$$

Proof. Let's consider an arbitrary type $E : \mathcal{U}$. There is a canonical function $c_{\sqcup}$ defined by

$$\begin{cases} (A \sqcup^C B \to E) & \longrightarrow & \mathsf{cocone}_{\mathscr{D}}(E) \\ \qquad\qquad t & \longmapsto & (t \circ \mathsf{inl}, t \circ \mathsf{inr}, \mathsf{ap}_t \circ \mathsf{glue}) \end{cases}$$

We write informally $t \mapsto t \circ c_{\sqcup}$ for this function. We show that this is an equivalence.

Firstly, given a $c = (i, j, h) : \mathsf{cocone}_{\mathscr{D}}(E)$, we need to construct a map $\mathsf{s}(c)$ from $A \sqcup^C B$ to E.

$$\begin{array}{ccc} C & \xrightarrow{g} & B \\ {\scriptstyle f}\downarrow & \swarrow^{h} & \downarrow{\scriptstyle j} \\ A & \xrightarrow[i]{} & E \end{array}$$

The map $\mathsf{s}(c)$ is defined in the following way

$$\begin{aligned} \mathsf{s}(c)(\mathsf{inl}(a)) &:\equiv i(a), \\ \mathsf{s}(c)(\mathsf{inr}(b)) &:\equiv j(b), \\ \mathsf{ap}_{\mathsf{s}(c)}(\mathsf{glue}(x)) &:= h(x). \end{aligned}$$

We have defined a map

$$\begin{cases} \mathsf{cocone}_{\mathscr{D}}(E) & \longrightarrow & (A \sqcup^C B \to E) \\ c & \longmapsto & \mathsf{s}(c) \end{cases}$$

and we need to prove that this map is an inverse to $t \mapsto t \circ c_{\sqcup}$. On the one hand, if $c = (i, j, h) : \mathsf{cocone}_{\mathscr{D}}(E)$, we have

$$\begin{aligned} \mathsf{s}(c) \circ c_{\sqcup} &= (\mathsf{s}(c) \circ \mathsf{inl}, \mathsf{s}(c) \circ \mathsf{inr}, \mathsf{ap}_{\mathsf{s}(c)} \circ \mathsf{glue}) \\ &= (\lambda a.\, \mathsf{s}(c)(\mathsf{inl}(a)),\ \lambda b.\, \mathsf{s}(c)(\mathsf{inr}(b)),\ \lambda x.\, \mathsf{ap}_{\mathsf{s}(c)}(\mathsf{glue}(x))) \\ &= (\lambda a.\, i(a),\ \lambda b.\, j(b),\ \lambda x.\, h(x)) \\ &\equiv (i, j, h) \\ &= c. \end{aligned}$$

On the other hand, if $t : A \sqcup^C B \to E$, we want to prove that $\mathsf{s}(t \circ c_{\sqcup}) = t$. For $a : A$, we have

$$\mathsf{s}(t \circ c_{\sqcup})(\mathsf{inl}(a)) = t(\mathsf{inl}(a))$$

because the first component of $t \circ c_{\sqcup}$ is $t \circ \mathsf{inl}$. In the same way, for $b : B$ we have

$$\mathsf{s}(t \circ c_{\sqcup})(\mathsf{inr}(b)) = t(\mathsf{inr}(b))$$

and for $x : C$ we have

$$\mathsf{ap}_{\mathsf{s}(t \circ c_{\sqcup})}(\mathsf{glue}(x)) = \mathsf{ap}_t(\mathsf{glue}(x))$$

hence $\mathsf{s}(t \circ c_{\sqcup}) = t$.

This proves that $c \mapsto \mathsf{s}(c)$ is a quasi-inverse to $t \mapsto t \circ c_{\sqcup}$, as desired. □